

Appendix 2.3: Regulations on Open Architecture, Domain Integration, and Agora Operations

1. Principles of Openness and Rejection of Monopoly

The Transceptor platform does not claim a monopoly on knowledge or an exclusive right to interpret socio-economic processes. It provides objective mathematical rules of interaction and a platform basis for integrating any domain fields—from science, medicine, and urban planning to culture and local municipal governance.

The open architecture of the platform guarantees the absence of centralized ideological censorship, ensuring equal access for citizens and institutions to rule-making tools and resource allocation. All procedural logic, interaction protocols, and audit algorithms operate on the basis of transparent smart contracts and open mathematical specifications.

2. Subjectivity Validation (IdentityValidator)

Access to voting processes, decision-making, and rule-making within the Digital Agora space is strictly regulated by the `IdentityValidator` software class. The class provides full protection against anonymous falsifications, Sybil attacks, and unauthorized interference by autonomous bot networks.

- **Identifier Verification:** The system performs mandatory validation of a verified decentralized identity profile (`did_hash`) and a cryptographically encrypted user signature.
- **Exception Handling:** In the absence of an attached DID profile or in case of an invalid cryptographic key, the `IdentityValidator` class instantly returns the `AUTH_ERR_NO_DID` exception, blocking any attempt to create an initiative or participate in voting.
- **Anonymity and Privacy:** Subjectivity validation is based on Zero-Knowledge proofs (ZK-KYC) and the principle of "Internal Blindness". This allows confirming the existence of civic rights and reputational status without disclosing personal biometric data, legal name, or physical location of the user.

3. Three-Layer Initiative Decompilation in Agora

Any civic or institutional initiative undergoes mandatory automated pre-moderation through a decentralized network of Semantic AI Oracles before being submitted to a public vote. Pre-moderation consists of a three-layer decompilation of the text and budget estimation of the project:

- **Layer 1: Jurisprudence ($Comp_{law}$)**

  Verification of the initiative text for compliance with the Fundamental Human Rights Contour, current legislative framework, and system invariants. Automated detection of hidden legal trapdoors, jurisprudential contradictions, and corruption loopholes is executed.
- **Layer 2: Budget Estimates ($Comp_{fin}$)**

  Vector parsing of all financial metrics and budgetary requirements using cosine similarity methods. This layer is designed to detect populist cost inflations, misuse of funds, and signs of cartel collusions or covert lobbying.
- **Layer 3: Ecology ($Comp_{eco}$)**

  Direct mathematical modeling of the comprehensive ecological footprint of the project. Carbon, acoustic, hydrological, and thermodynamic parameters of impact on the living environment are calculated.

3.1. Cognitive Mediation Protocol (AI-Architect Engine)

Threshold Admission Filter:

The final admission of an initiative to the general voting stage is calculated via the Heaviside step function across all three layers:

$$\Psi = \text{Step}(\text{Comp}*{\text{law}} - \text{Th}*{\text{law}}) \cdot \text{Step}(\text{Comp}*{\text{fin}} - \text{Th}*{\text{fin}}) \cdot \text{Step}(\text{Comp}*{\text{eco}} - \text{Th}*{\text{eco}})$$

where $\text{Th}*{\text{law}}$, $\text{Th}*{\text{fin}}$, and Th_{eco} are strictly established threshold values of the corresponding standards.

- If $\Psi = 1$, the initiative is recognized as valid and transferred to the voting contour of citizens and Personal Companions.
- If $\Psi = 0$, at a zero value of the threshold filter ($\Psi = 0$), a civil or institutional initiative is not automatically rejected, but transferred to the AI-Architect auto-completion loop. The AI-Architect Engine module cleanses the proposal of emotional syntax, performs automatic vector modeling of the legal, financial, and ecological layers, and returns a ready ScenarioBrief to the author for one-click approval and re-submission to the Agora.

4. Dual-Contour Isonomy and Governance Architecture (Contour Boundary Enforcer)

4.1. Value Contour (Absolute Isonomy)

- **Absolute Equality Invariant:** In the Value Contour, which regulates fundamental human rights, ethics, culture, environmental principles, and strategic humanitarian orientation, absolute equality of citizens is guaranteed on the principle of "1 person = 1 vote".
- **Effective Electoral Weight:** The effective voting weight of any resident is strictly fixed at $W_{effective} = 1.0$.
- **Prohibition of Capital and Meritocratic Surcharges:** All meritocratic multipliers (W_{user}), accumulated reputation indices ($\Delta I_{sc}, \Delta I_{cp}, \Delta I_{ft}$), and financial assets are forcibly zeroed out when constructing a Value Contour transaction. Corporate entities, legal persons, and commercial pools are hardware-prohibited from voting ($W_{effective} = 0$).
- **Prohibition of Quadratic Voting:** The use of Quadratic Voting (QV) and the expenditure of T_{net} tokens in the Value Contour is strictly forbidden.
- **Auto-clearing Fraud Penalties:** Any attempt to initiate quadratic voting or exert financial influence in the Value Contour is classified as architectural fraud, triggering an immediate cascade reduction of the offender's meritocratic weight to the technological minimum ($W_{user} = 0.01$).

4.2. Mathematical Contour (Dynamic Meritocracy)

- **Dynamic Electoral Weight:** The Mathematical Contour regulates budget allocation, cost estimation, RWA infrastructure projects, equipment procurement, and logistics. Voting weight $W_{user} \in [0.01; 5.00]$ is calculated dynamically based on confirmed Proof-of-Contribution indices (intellectual ΔI_{cp}, social ΔI_{sc}, and fiscal ΔI_{ft}).
- **Quadratic Voting Protocol (QV):** Decision-making and procurement of additional electoral weight are executed through an exponential barrier:

  - $C(V) = V^2$

  where V is the number of additional votes purchased, and $C(V)$ is the cost in T_{net} tokens. The total cost of N votes is calculated as:

  - $C_{total}(N) = \sum_{V=1}^{N} \quad V^2$

- **Anti-Lobbying Barrier:** The quadratic cost scaling renders commercial pool collusion and narrow economic lobbying mathematically ruinous.

4.3. Contour Boundary Enforcer (Atomic Proposal Split)

- **Attempted Concealment Trigger:** If a project author attempts to disguise a budget-intensive economic proposal under the guise of a humanitarian or ethical manifesto, the Contour Boundary Enforcer module executes an automatic Atomic Proposal Split:
  1. Humanitarian/Ethical Manifesto → Routed to Value Contour ($W_{effective} = 1.0$, No QV).
  2. Executive Feasibility Study and Budget Estimate → Routed to Mathematical Contour ($W_{user} \in [0.01; 5.00]$, $C(V) = V^2$, 3-layer ScenarioBrief).
- **Dual Approval Requirement:** Financial execution and escrow release are unlocked strictly upon the successful approval of both linked acts.

4.4. Anti-Collusion and Guild Protection (Double-Blind Rotation & Style Neutralizer)

- **NLP Style Sanitizer:** Mandatory automated stripping of stylistic quirks, jargon, formatting, and author-identifying syntax from project evaluation reports prior to expert review.
- **Double-Blind Commit-Reveal:** Complete mutual anonymization of experts and authors. Evaluation scores and audit verdicts are submitted via zero-knowledge commitments (ZK-Commitments).
- **Guild Expert Rotation:** Randomized dynamic rotation of expert panels to eliminate cronyism, nepotism, and expert cartel agreements.